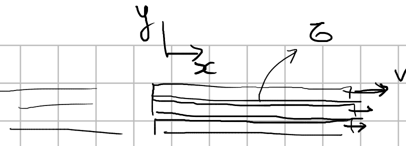
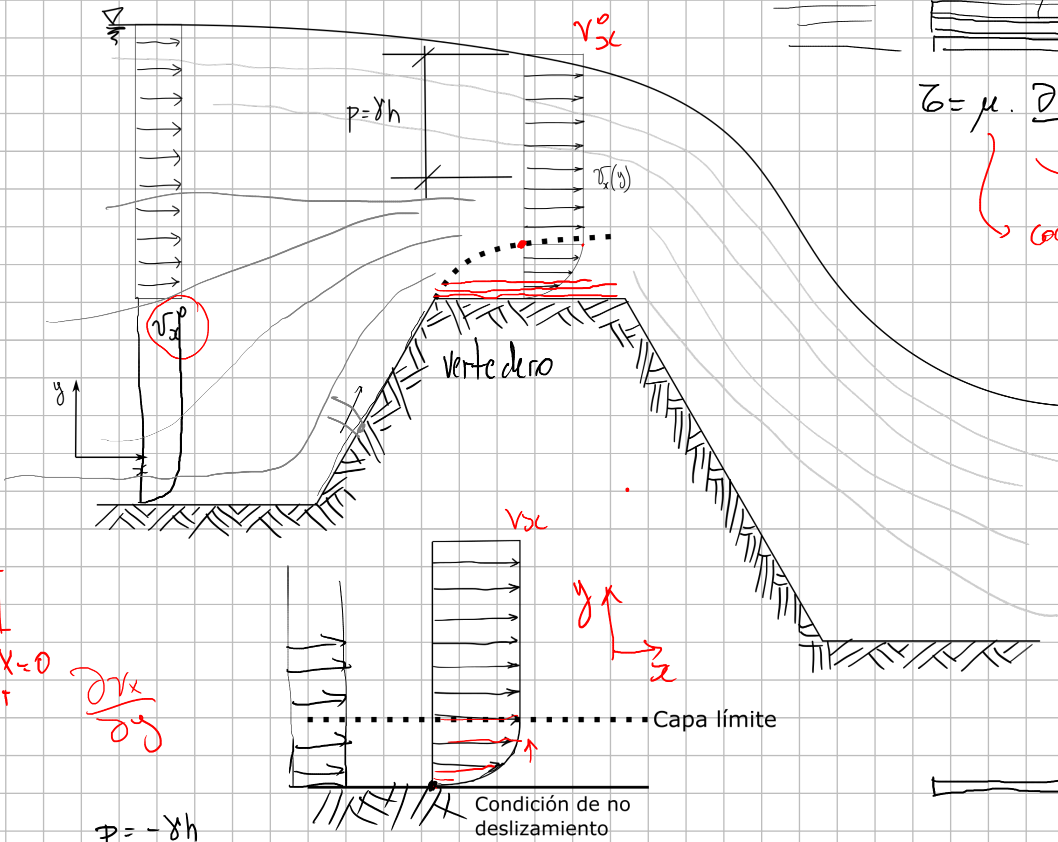
Fluidos Viscosos



$$\tau = \mu \cdot \frac{\partial v_x}{\partial y}$$

Ley de Newton de fluido viscoso

coeficiente de viscosidad



$$\begin{cases} \nabla \cdot \underline{\sigma} + X = \rho a \\ \underline{\sigma} = \underline{\sigma}^T \end{cases}$$

$$\begin{cases} \nabla \cdot \underline{\sigma} + X = 0 \\ \underline{\sigma} = \underline{\sigma}^T \end{cases}$$

$$\frac{\partial v_x}{\partial y}$$

$$\frac{\partial v_y}{\partial x} \qquad \frac{\partial v_x}{\partial y}$$

$$p = -\gamma h$$

$$\sigma_{xx} = p \qquad \sigma_{yy} = p$$

$$+ \qquad \tau_{xy} = \tau_{yx} \quad \text{viscosidad}$$

$$\tau_{xy} = \mu \cdot \frac{\partial v_x}{\partial y}$$

$$\boxed{\sigma_{ij} = \sigma_{ji}}$$

$$\sigma_{12} = \sigma_{21}$$

$$\sigma_{ij} = -\sigma_{ji} \Rightarrow$$

$$\underline{\sigma} = \begin{bmatrix} p & \mu \cdot \frac{\partial v_x}{\partial y} \\ \mu \cdot \frac{\partial v_x}{\partial y} & p \end{bmatrix}$$

$$\underline{\sigma} = \begin{bmatrix} \overset{-\gamma h}{p} & \mu \cdot \frac{\partial v_x}{\partial y} & 0 \\ \mu \cdot \frac{\partial v_x}{\partial y} & p & 0 \\ 0 & 0 & p \end{bmatrix}$$

Fluido newtoniano incompresible

con $i \neq j$

$$\sigma_{ij} = 2\mu \underbrace{V_{ij}}_{\text{tasa de deformación}} = 2\mu \cdot \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)$$

$$\sigma_{xy} = \mu \cdot \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}\right) = \mu \cdot \frac{\partial v_x}{\partial y}$$

$$\sigma_{yx} = \mu \cdot \left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y}\right) = \mu \cdot \frac{\partial v_x}{\partial y}$$

$$\Sigma M = 0$$

$$\Sigma M = \tau_{xy} \cdot h - \tau_{yx} \cdot h = 0$$

$$(\tau_{xy} - \tau_{yx}) \cdot h = 0 \qquad h \to 0 \neq 0$$

$$\tau_{xy} = \tau_{yx}$$

α

$$\underline{\underline{\Gamma}} = \begin{bmatrix} p & \mu\cdot\frac{\partial v_x}{\partial y} \\ \mu\cdot\frac{\partial v_x}{\partial y} & p \end{bmatrix}$$

$p \neq 0$ y $\alpha \neq 0$

$$\underline{n} = \begin{bmatrix} \cos\theta \\ \operatorname{sen}\theta \end{bmatrix}$$

Vector tensión

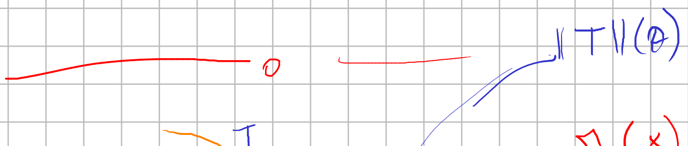
$$\underline{T} = \underline{\underline{\Gamma}}\cdot\underline{n} = \begin{bmatrix} p & \alpha \\ \alpha & p \end{bmatrix}\cdot\begin{bmatrix} n_x \\ n_y \end{bmatrix} = \begin{bmatrix} p\,n_x + \alpha\,n_y \\ \alpha\,n_x + p\,n_y \end{bmatrix}$$

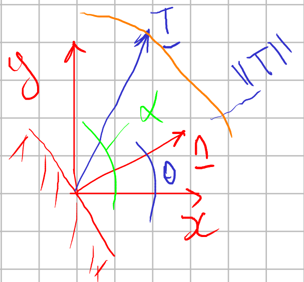
Hidroestático

$$\|T\|^2 = |p|^2$$

$$\|\underline{T}\|^2 = (p n_x + \alpha n_y)^2 + (\alpha n_x + p n_y)^2$$
$$= (p n_x)^2 + 2\,p n_x\,\alpha n_y + (\alpha n_y)^2 +$$
$$(\alpha n_x)^2 + 2\,\alpha n_x \cdot p n_y + (p n_y)^2 =$$
$$\|\underline{T}\| = \sqrt{p^2 + \alpha^2 + 2\,\alpha p\,\operatorname{sen}(2\theta)}$$

$$\alpha = 0 \longrightarrow \frac{\partial v_x}{\partial y} = 0$$





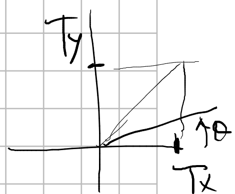
$\underline{\underline{\Gamma}}(\underline{x})$

$$\underline{T} = \underline{\underline{\Gamma}}\cdot\underline{n}$$

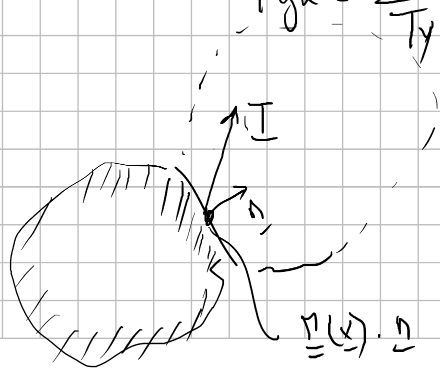
$$\underline{n} = \begin{bmatrix} \cos\theta \\ \operatorname{sen}\theta \end{bmatrix}$$

$$\|\underline{n}\| = 1$$

$$\begin{bmatrix} T_x \\ T_y \end{bmatrix} = \underline{T} = \underline{\underline{\Gamma}}\cdot\underline{n} = \begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{xy} & \sigma_y \end{bmatrix}\cdot\begin{bmatrix} n_x \\ n_y \end{bmatrix} = \begin{bmatrix} \sigma_x\cdot\cos\theta + \tau_{xy}\cdot\operatorname{sen}\theta \\ \tau_{xy}\cdot\cos\theta + \sigma_y\cdot\operatorname{sen}\theta \end{bmatrix}$$



$$\operatorname{tg}\alpha = \frac{T_x}{T_y} \longrightarrow \theta(\alpha) \longrightarrow \|\underline{T}\|(\alpha)$$



$$\underline{\Gamma} = \begin{bmatrix} T_x(\theta) \\ T_y(\theta) \end{bmatrix}$$

$$\underline{\Gamma}(\theta) = \begin{bmatrix} \Gamma_x(\theta) \\ \Gamma_y(\theta) \end{bmatrix}$$